

SIH Project: Section 479 Undertrial Release & Prison Rehabilitation System

One-line PS: A digital pipeline to identify, fast-track, and track Section-479-eligible undertrial prisoners through to actual release — and connect in-custody rehabilitation to real post-release employment — to reduce prison overcrowding in India.

Core thesis: Overcrowding is largely a *visibility* problem, not just a capacity problem. A large share of "overcrowded" prisoners are already legally entitled to release; they're stuck in an invisible paperwork lag. Make the pipeline visible end-to-end, and legitimate releases happen on time.

How to use this doc

Each module below lists: what it tracks, what real-world system/portal it maps to (for grounding your pitch), a basic MVP approach your team can actually build in hackathon time, and suggested skillset. Assign one module (or a pair of related modules) per teammate.

MODULE A — Eligibility & Exclusion Engine

What it tracks: Which undertrials currently qualify for release under Section 479.

Logic (this is the legal rule, get it right):

Eligible if time in custody $\geq \frac{1}{2}$ of max sentence for the charged offence

Eligible if time in custody $\geq \frac{1}{3}$ of max sentence AND first-time offender (no prior conviction)

Excluded if offence carries death penalty or life imprisonment

Excluded if the person has more than one pending case (investigation/inquiry/trial)

Real-world equivalent: State jail management records (e-Prisons MIS)

Basic MVP solution: A rules-based calculator — no ML needed. Input: custody start date, offence, max sentence, prior-conviction flag, pending-case count. Output: eligible / not eligible / excluded (with reason). This is pure date-math and conditionals.

Skillset: 1 person — backend/logic, no special expertise needed.

MODULE B — Application / Paperwork Auto-Generation

What it tracks: Which flagged prisoners need a bail/personal bond application drafted.

Real-world equivalent: Standard bail/bond application formats used by DLSA panel lawyers.

Basic MVP solution: A document template with placeholder fields (name, offence, custody dates, sentence slab, prior record) auto-filled from Module A's output, exported as PDF.

Skillset: 1 person — templating/document generation.

MODULE C — Court Filing & Status Tracking (the "faster system" loop)

What it tracks: Once filed — CNR number, hearing date, order outcome (bail granted/denied), and pushing that outcome back to the jail immediately.

Real-world equivalent: eCourts Services Portal (services.ecourts.gov.in) — India's official, free, public case-status lookup by CNR number. This is the actual system that currently does NOT feed back into prison records automatically — that gap is your differentiator.

Basic MVP solution: For the demo, simulate this — a mock "court response" you trigger manually or on a timer (flagged → filed → hearing scheduled → order passed). State clearly in your pitch that a production version would poll the eCourts case-status API by CNR; for the hackathon you're demonstrating the workflow, not live-scraping government servers.

Important boundary: This system does NOT decide bail. It only fast-tracks filing and closes the notification gap after the judge decides. Say this explicitly in your pitch — judges will ask.

Skillset: 1-2 people — integration/mock API, workflow state machine.

MODULE D — Legal Aid & Bond/Surety Assistance

What it tracks: DLSA panel lawyer assignment; whether the bond/surety condition (the #1 reason people stay in jail even after winning bail) has been met.

Real-world equivalent: District Legal Services Authority (DLSA) / NALSA legal aid workflow; State Legal Services Authority financial assistance schemes for indigent undertrials.

Basic MVP solution: A simple assignment queue (round-robin or district-based) + a checklist status (bond amount, surety arranged y/n) with a stall alert if it's been stuck too long.

Skillset: 1 person — backend/simple workflow logic.

MODULE E — Overcrowding Dashboard & Capacity Prediction

What it tracks: Current occupancy vs. sanctioned capacity per jail, staff-to-inmate ratio, and a forward projection of occupancy.

Real-world equivalent: NPIP (National Prisons Information Portal) publishes aggregate, after-the-fact stats. Nobody currently projects forward.

Basic MVP solution:

Live view: occupancy vs. capacity per (mock) jail, color-coded by severity

Prediction: simple date-math — project expected releases using known sentence-end dates + Module A's eligibility dates, minus expected new admissions (use a historical average rate). No ML needed for MVP; mention a time-series model as a stretch goal.

Bonus: split out "how much of this jail's overcrowding is eligible-but-not-yet-processed" vs. genuine capacity load — ties directly back to Module A/C data and is a strong pitch point.

Skillset: 1-2 people — frontend/data visualization.

MODULE F — Rehabilitation Tracking

What it tracks: Enrollment and progress in vocational/education programs; certification status.

Real-world equivalent: State programs already exist — Haryana's ITI/polytechnic courses in jails, Tihar's DSEU certification tie-up, Assam's vocational programs. All are state-specific and don't currently follow the inmate across transfers or to release.

Basic MVP solution: An enrollment + progress tracker (percentage complete per course) with an auto-generated digital certificate once a course is marked complete. This certificate record is what Module G consumes later.

Skillset: 1 person — backend/CRUD, simple UI.

MODULE G — Market Linkage & Post-Release Job Placement

What it tracks: (a) linking in-custody production/work to an existing marketplace, (b) matching a released inmate's certified skills to local employers/NGOs/self-employment schemes.

Real-world equivalent:

In-custody: Kara Bazaar (existing govt portal selling inmate-made goods) — don't rebuild this, position your system as feeding into it

Post-release: scattered state schemes (e.g., West Bengal's Swanirbhar Karmasansthan Prakaalpa) — no national matching layer exists. This is a genuine, clearly-missing piece — good to lead with in your pitch.

Basic MVP solution:

In-custody: a simple "connect to Kara Bazaar" link/mock listing flow tied to Module F's tracked skills

Post-release: a basic skill-tag matching table — filter a mock employer/NGO directory by skill + district, surface top matches at release time (this is a "digital skill passport" concept)

Skillset: 1-2 people — backend + basic matching logic.

MODULE H — Notifications

What it tracks: Alerts to family/next-of-kin (case status, hearing dates), jail staff (stalled applications), DLSA.

Basic MVP solution: Use a free-tier SMS/WhatsApp sandbox (e.g., Twilio trial) for a live demo, or a simple in-app notification log if you want to avoid third-party signup overhead.

Skillset: 1 person — can be combined with another module if team is small.

MODULE I — Compliance Reporting (bonus, low effort / high pitch value)

What it tracks: The exact numbers states/UTs are already required to report to the Supreme Court — number of eligible undertrials identified, applications filed, releases completed.

Basic MVP solution: This is nearly free once Modules A–C exist — just an aggregation/export view (PDF or Excel) of data you're already tracking. Strong "real government use-case" talking point for judges.

Skillset: Whoever finishes their module first.

What NOT to build (state this explicitly in your pitch — it shows judgment)

No automated bail *approval* — the judge always decides; the system only speeds up filing and closes the notification gap

No live scraping of eCourts/e-Prisons in the actual hackathon build — mock this integration and state the real path as future scope

No "risk of reoffending" / recidivism prediction scoring — ethically and legally fraught (bias, due process); keep AI components to identification and workflow only, never to release decisions

Data for your demo

No public individual-level prisoner data exists (and shouldn't — it's private data). Build a synthetic dataset: ~50–100 mock prisoner records with offence, max sentence, custody start date, prior-conviction flag, pending-case count.

For real aggregate numbers to cite on slides: NCRB's "Prison Statistics India" annual report, the NPIP portal, and Rajya Sabha replies from the Ministry of Home Affairs on Section 479 implementation.

Suggested team split (adjust to your actual team size)

Eligibility + paperwork lead — Modules A + B

Court/tracking + legal aid lead — Modules C + D

Dashboard/data lead — Module E

Rehab + placement lead — Modules F + G

Notifications + reporting — Modules H + I

Research/pitch lead — verifies legal facts (Section 479 text, UTRC process), builds the 9-slide narrative, sources the real statistics for slides